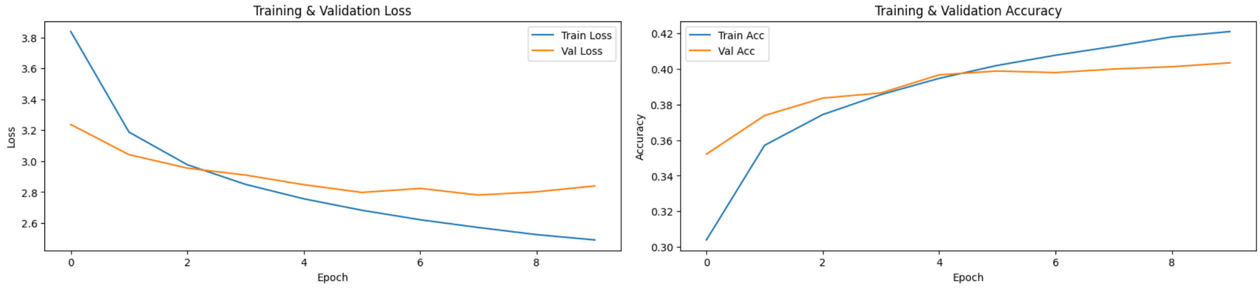
Practical Work 3: Image Captioning

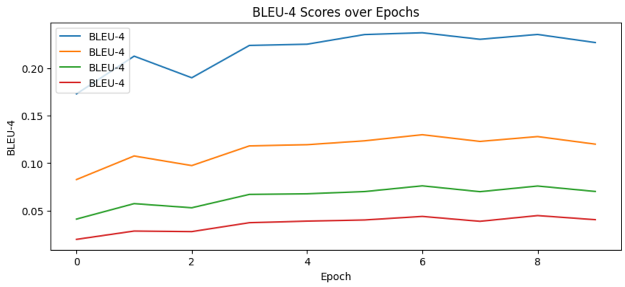
For image preprocessing, we used separate transforms for training and testing. Training images are converted to RGB, resized, randomly cropped to 224×224, randomly flipped, color-jittered, converted to tensors, and normalized with ImageNet statistics. Test images use a deterministic pipeline: RGB conversion, resizing to 224×224, tensor conversion, and ImageNet normalization.

Model 1 - Show and Tell

Model 1 used a ResNet-18 encoder with the final fully connected layer removed, producing 512-dimensional features, and an LSTM decoder with 256-dim word embeddings and 512 hidden units.



Training ran for 10 epochs with dropout 0.5 and Adam at 0.001. Losses dropped steadily (train from 3.8 → 2.48, validation from 3.2 → ~2.81), while accuracy rose to ~42% on training and ~40% on validation, showing the model learned word patterns and sentence structure.



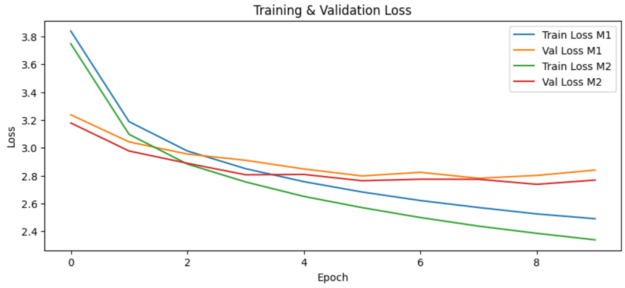
BLEU-1 to BLEU-4 scores improved modestly, with BLEU-1 around 0.23 and BLEU-4 close to 0.04, reflecting limited overlap with reference captions.
Here are some examples of generated captions:

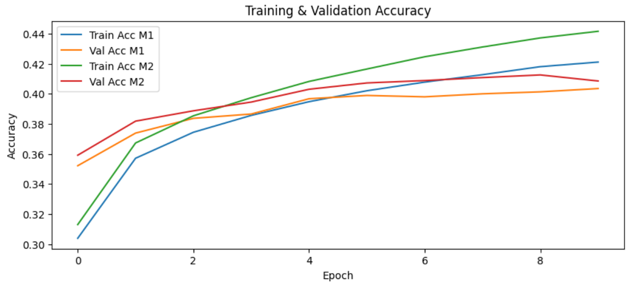
```
---
Reference: A boy smiles in front of a stony wall in a city .
Generated: a man in a blue shirt is standing on a bench
---
Reference: A little boy at a lake watching a duck .
Generated: a man in a black shirt is standing on a pier near a lake
---
Reference: A man uses ice picks and crampons to scale ice .
Generated: a child in a red jacket is snowboarding
---
```

Captions are grammatically correct and fluent but often generic, defaulting to “a man...” and failing to capture specific details like the duck.
Overall, the model trained smoothly and produced plausible sentences, but its main limitation was weak grounding in the image content and low semantic accuracy.
We doubt that further training with these hyperparameters would improve the results to much as looking at the graphs makes us think that it will probably go into overfitting.

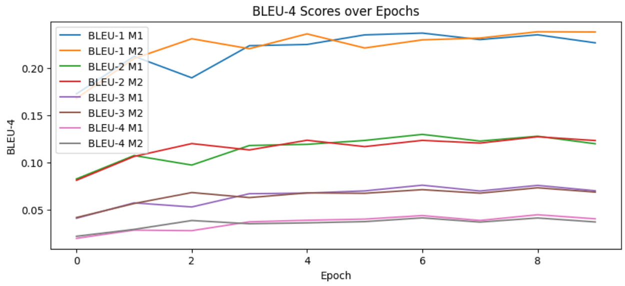
Model 2 - Show, Attend and Tell

We kept the hyperparameters the same in the two models to have more consistent results. The difference between this model and the first one is the addition of the attention mechanism with 256 dimensions.





Losses dropped steadily, with training loss decreasing from 3.75 to 2.34 and validation loss from 3.18 to about 2.77. Accuracy improved to around 44% on training and 41% on validation, showing that the model learned caption structure slightly better than Model 1.



BLEU scores improved modestly, with BLEU-1 around 0.24 and BLEU-4 around 0.04, which is very similar to Model 1 and still shows limited overlap with the reference captions.

Here are some examples of generated captions:

```
---
Reference: A young child is walking on a stone paved street with a metal pole and a man behind him .
Generated: a man in a blue shirt is standing on a sidewalk
---
Reference: A child and a woman are at waters edge in a big city .
Generated: a man in a black shirt and a white shirt is standing in the water
---
Reference: A man uses ice picks and crampons to scale ice .
Generated: a man in a red jacket is skiing in a snow slope
---
```

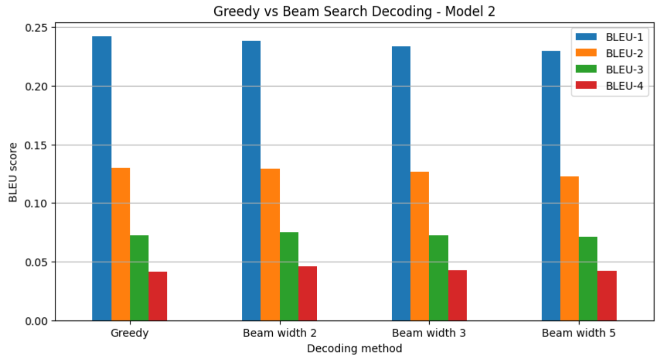
Compared to model 1, model 2 seems to be doing similar errors and also always defaults to “a man”. Same with the first one we think that more training could probably lead to very slight increases but then very quickly overfit.

Additional Task 1: Beam Search

Hypothesis

Beam search will probably improve BLEU scores because it considers several candidate caption sequences instead of only one. Increasing the beam width too much may produce captions that are too generic.

Conclusion



The hypothesis was confirmed as beam 2 achieved results better results for BLEU-3 and BLEU-4 but the greedy approach has still the best results for BLEU-1 and BLEU-2. Increasing the beam width gets progressively worse results as beam 3 has very similar results to greedy for BLEU-3 and BLEU-4 and beam 5 has all worse results.

Additional Task 2: Attention Visualization

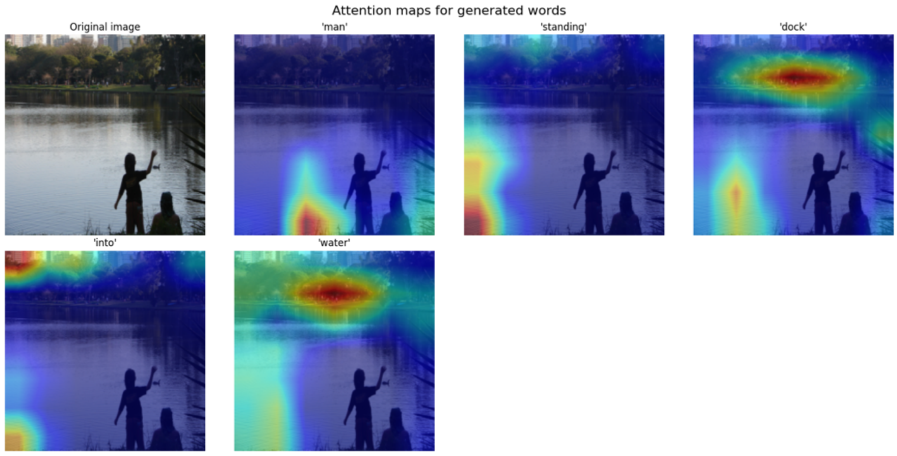
Hypothesis

The regions more relevant to the generated words will probably be given more attention. For example, for words involving humans like "man" or "child", we expect the human subject of the photo to be given more attention, and for words more oriented to the scene the attention will probably more spread.

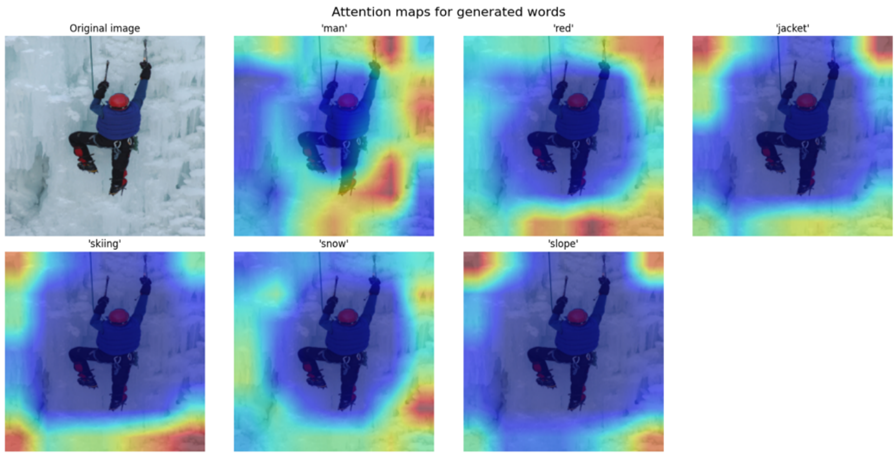
Conclusions

The accuracy of the attention seems to vary depending by the word.

For example, the attentions for the word "man" and "blue" seem to be accurate in first image of the child, but in the same image for the word "shirt" the attention is completely off.
In the other two images the attention is off even for the word "man".



For the second image of the lake, the attention for "water" is completely off but for "dock" it could be understandable.



For the last image of the ice climber the attention seems to be almost always to be around the man independent of the word, even when the word is "man", "red" or "jacket" which should be directly on the climber.
Looking at these attention maps we could say that the model somewhat learned some visual grounding, but the attention remains spread rather than focused on the object of the generated word.